

Selected Topics Of Living Systems HW3

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Technion

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1 Hydrodynamics

1. The Maxwell model for viscoelastic fluids corresponds to a spring and dashpot in series. Its rheology is given by

$$(1 + \tau \partial_t) \tilde{\sigma} = \eta \dot{\gamma} \quad (1)$$

where $\tau = \eta/G$ is the relaxation time, $\tilde{\sigma}$ is the shear stress, η is the viscosity and $\dot{\gamma}$ is the shear strain rate. Show explicitly that the equation is solved by

$$\tilde{\sigma} = G \int_0^t dt' \dot{\gamma}(t') e^{-(t-t')/\tau} \quad (2)$$

demonstrate the solid/fluid behaviors for $t \ll \tau$ and $t \gg \tau$.

2. The Kelvin-Voigt model corresponds to a spring and dashpot in parallel. What is the rheology of this model ($\sigma(\gamma, \dot{\gamma})$)? How does this material behave at long times?
3. Consider a rigid body rotation, whose velocity field is described by $\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r}$ with $\boldsymbol{\Omega}$ the angular velocity. Show that the tensor $\partial_\alpha v_\beta$ is antisymmetric in this case.

1.1 Maxwell model for viscoelastic fluids

inserting $\eta = \tau G$ and the given integral expression

$$(1 + \tau \partial_t) \int_0^t dt' \dot{\gamma}(t') e^{-(t-t')/\tau} = \tau \dot{\gamma} \quad (3)$$

pulling a constant w.r.t the integral out

$$(1 + \tau \partial_t) e^{-t/\tau} \int_0^t dt' \dot{\gamma}(t') e^{t'/\tau} = \tau \dot{\gamma} \quad (4)$$

If we designate $A(t) = e^{-t/\tau}$ and $B(t) = \int_0^t dt' \dot{\gamma}(t') e^{t'/\tau}$ then the equation becomes

$$(1 + \tau \partial_t) A(t) B(t) = A(t) B(t) + \tau \left(\frac{dA(t)}{dt} B(t) + A(t) \frac{dB(t)}{dt} \right) \quad (5)$$

and since

$$\frac{dA}{dt} = -\frac{1}{\tau} e^{-t/\tau} = -\frac{1}{\tau} A(t) \quad (6)$$

we can rewrite

$$(1 + \tau \partial_t) A(t) B(t) = A(t) B(t) + \tau \left(-\frac{1}{\tau} A(t) B(t) + A(t) \frac{dB(t)}{dt} \right) \quad (7)$$

$$= A(t) \left(B(t) + \tau \left(-\frac{1}{\tau} B(t) + \frac{dB(t)}{dt} \right) \right) \quad (8)$$

$$= \tau A(t) \frac{dB(t)}{dt} \quad (9)$$

inserting into equation 3

$$e^{-t/\tau} \partial_t \int_0^t dt' \dot{\gamma}(t') e^{t'/\tau} = \dot{\gamma} \quad (10)$$

but

$$\partial_t \int_0^t dt' \dot{\gamma}(t') e^{t'/\tau} = \dot{\gamma}(t) e^{t/\tau} \quad (11)$$

so eventually

$$\dot{\gamma} = \dot{\gamma} \quad (12)$$

so the expression for $\tilde{\sigma}$ solves the equation.

1.1.1 $t \ll \tau$

In this limit

$$\tilde{\sigma} = G \int_0^t dt' \dot{\gamma}(t') e^{-(t-t')/\tau} \simeq G \int_0^t dt' \dot{\gamma}(t') = G(\gamma(t) - \gamma(0)) \quad (13)$$

and since the stress is linear in strain, as we know from the lectures, the system behaves as an ideal elastic solid.

1.1.2 $t \gg \tau$

In this limit the exponential term in the integral expression for $\tilde{\sigma}$ decays quickly and earlier values of $\dot{\gamma}$ are cancelled out by the exponential approaching zero. This means we can assume the main contribution comes from very late times so we can treat $\dot{\gamma}$ as a constant:

$$\tilde{\sigma} = G \int_0^t dt' \dot{\gamma}(t') e^{-(t-t')/\tau} \simeq G \dot{\gamma}(t) \int_0^t dt' e^{-(t-t')/\tau} \quad (14)$$

$$= \tau G \dot{\gamma}(t) \quad (15)$$

$$= \eta \dot{\gamma} \quad (16)$$

which is the equation for an ideal viscous liquid as we know from the lectures.

1.2 Kelvin-Voigt model

Since the spring and dashpot are connected in parallel, the strain on both of the elements will be the same, the strain comes from two separate contributions:

$$\gamma = \gamma_{spring} = \gamma_{dashpot} \quad \sigma = \sigma_{spring} + \sigma_{dashpot} \quad (17)$$

as we know from the lectures

$$\sigma_{spring} = G\gamma \quad \sigma_{dashpot} = \eta \dot{\gamma} \quad (18)$$

which gives us

$$\sigma = G\gamma + \eta \dot{\gamma} \quad (19)$$

which is the Kelvin-Voigt rheology. We can treat this as a separable differential equation in γ whose solution is

$$\gamma = \frac{\sigma_0}{G} + ce^{-Gt/\eta} \quad (20)$$

where σ_0 is some stress value. For very large times

$$\gamma = \frac{\sigma_0}{G} \quad (21)$$

1.3 Antisymmetric tensor

I'll rewrite the velocity using levi-civita symbols

$$v_i = \varepsilon_{ijk} \Omega_j r_k \quad (22)$$

applying the derivative

$$\partial_\alpha v_\beta = \varepsilon_{\beta jk} \Omega_j (\partial_\alpha r_k) = \varepsilon_{\beta jk} \Omega_j \delta_{\alpha k} = \varepsilon_{\beta j\alpha} \Omega_j = -\varepsilon_{\alpha j\beta} \Omega_j = -\partial_\beta v_\alpha \quad (23)$$

therefore it's antisymmetric.

2 Maxwell model in dividing tissues

Consider a dividing tissue, where the cellular density satisfies

$$\partial_t \rho = -\partial_\alpha (\rho v_\alpha) - K(\rho) \quad (24)$$

the reaction term on RHS describes division and death. Suppose that close to the homeostatic (steady-state) density ρ_h , the reaction is given by $K(\rho) \approx \frac{1}{\tau} (\rho - \rho_h)$ with τ a typical density relaxation time due to division and death. In addition, assume that the pressure around the homeostatic pressure P_h satisfies to linear order $P - P_h = K(\rho - \rho_h)$. Denote $\delta\sigma = P_h - P$. Linearize the cellular density equation around the homeostatic state and show that $\delta\sigma$ behaves like a Maxwell model. What is the viscosity in this case? Note that this is a very simplified model that neglects, for example, cell-cell interactions.

Since we are near the homeostatic state we can insert the relations given

$$\partial_t \rho = -\partial_\alpha (\rho v_\alpha) - \frac{1}{\tau} (\rho - \rho_h) \quad (25)$$

We are looking at states near the homeostatic, so we can replace $\rho = \rho_h + \delta\rho(t)$ therefore

$$\partial_t (\delta\rho(t)) = -\partial_\alpha ((\rho_h + \delta\rho) v_\alpha) - \frac{\delta\rho(t)}{\tau} \quad (26)$$

$$= -\rho_h \partial_\alpha v_\alpha - \partial_\alpha (\delta\rho(t) v_\alpha) - \frac{\delta\rho(t)}{\tau} \quad (27)$$

The 2nd element in equation 27 contains two very small elements since we're near the homeostatic state so we can consider it negligible

$$\partial_t \delta\rho(t) = -\rho_h \partial_\alpha v_\alpha - \frac{\delta\rho(t)}{\tau} \quad (28)$$

next we notice that

$$\delta\rho(t) = \frac{P - P_h}{K} = -\frac{\delta\sigma}{K} \quad (29)$$

inserting this

$$\frac{1}{K} \partial_t \delta\sigma = \rho_h \partial_\alpha v_\alpha - \frac{\delta\sigma}{\tau K} \quad (30)$$

rearranging

$$(1 + \tau \partial_t) \delta\sigma = \tau \rho_h K \partial_\alpha v_\alpha \quad (31)$$

Which is the same structure as the Maxwell model. We can identify $\tau \rho_h K$ as the viscosity.